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Sleeve mounted onto a low-pressure shaft in a turbomachine

Abstract

The present invention relates to a rotating assembly of a turbomachine, comprising: —a shaft configured to drive a propeller of the turbo machine around an axis and comprising a first flange radially extending from the shaft, the first flange being monolithic with the shaft; —a sleeve fitted onto the shaft and comprising a second flange radially extending from the sleeve, the sleeve being attached to the first flange of the shaft by means of the second flange; and —at least one component connected to the shaft by means of the sleeve so as to be rigidly connected to the shaft to move therewith, the component comprising at least one sealing element.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to the field of aeronautical propulsion systems and finds in particular an application in the propulsion systems having a high bypass ratio.

BACKGROUND

(2) In a manner known per se, a turbomachine comprises bearings configured to support a mobile part relative to a fixed part (or stator) of the turbomachine or relative to another mobile part. These bearings can in particular comprise an inner ring mounted on a mobile element, for example a shaft, and an outer ring mounted on another part of the turbomachine. The inner ring can be particularly mounted on the low-pressure shaft using threaded nuts. The shaft then comprises complementary threads allowing the nut to be fixed and tightened against the inner ring. Sealing elements such as knife-edges can also be provided on the shaft. These sealing elements can be particularly monolithic with the low-pressure shaft.

(3) The mounting and machining of these different components however requires specific machining such as making threads or sealing elements which, in addition to raising technical difficulties in terms of machining, form privileged areas of damage of the shaft, creating areas of weakness and/or areas of stress concentration. Depending on the materials chosen for the shaft and the knife-edge-type sealing elements, there may also be increased risks of corrosion, in particular at the top of the knife-edges due to friction, this area cannot be protected.

(4) The maintenance operations are further complex, particularly when the worn or damaged areas touch the shaft.

(5) These risks are all the more significant in the case of supercritical and fast low-pressure shafts, for example in the turbomachines whose fan is unducted.

SUMMARY

(6) One aim is to overcome the aforementioned drawbacks, by proposing a low-pressure turbomachine shaft that is simpler to make, has an increased service life and whose maintenance is facilitated.

(7) For this purpose, according to a first aspect, a rotating assembly in accordance with claim 1 is proposed. Embodiments of the rotating assembly are defined in the dependent claims.

(8) According to a second aspect, a rotating assembly is proposed comprising: a shaft configured to drive a propeller of the turbomachine about an axis and comprising a first flange extending radially from the shaft, the first flange being monolithic with the shaft; a sleeve fitted onto the shaft and comprising a second flange extending radially from the sleeve, the sleeve being fixed onto the first flange of the shaft via the second flange; and at least one component connected to the shaft via the sleeve so as to be secured in movement to the shaft, the component comprising at least one sealing element.

(9) Some preferred but non-limiting characteristics of the rotating assembly according to the second aspect are as follows, taken individually or in combination: the second flange is monolithic with the sleeve; the rotating assembly further comprises a fixing member mounted on the sleeve, the at least one component being tightly mounted on the sleeve via the fixing member; the sleeve further comprises a shoulder extending at a distance from the second flange and the at least one component is tightly mounted between the fixing member and the shoulder; the at least one component is removably mounted on the sleeve and further comprises the at least one of the following elements: a ring of a bearing; and a scoop connected to a lubrication system. the component comprises two sealing elements delimiting an enclosure, an inner ring of a bearing housed in the enclosure and a scoop, the scoop comprising an open orifice on the enclosure and a lubrication fluid supply channel fluidly connecting the orifice to the bearing in order to supply the bearing with lubrication fluid; the first flange is further connected to an additional component of the turbomachine, the additional component comprising at least one sealing element, the sealing element preferably comprising a sealing knife-edge; the first flange is further connected to an additional component of the turbomachine, the additional component comprising an additional flange extending radially from the fixing member; and/or the fixing member is radially outside relative to the sleeve.

(10) According to a third aspect, it is proposed a turbomachine comprising a rotating assembly according to the first or the second aspect. The turbomachine can further comprise an unducted

propeller driven in rotation by the shaft. In one embodiment, the turbomachine has a bypass ratio greater than 30.

(11) According to a fourth aspect, the invention proposed an aircraft comprising at least one turbomachine according to the third aspect.

Description

DESCRIPTION OF THE FIGURES

(1) Other characteristics, aims and advantages will emerge from the following description, which is purely illustrative and not limiting, and which should be read in relation to the appended drawings in which:

(2) FIG. 1 schematically illustrates an example of a rotating assembly in accordance with one embodiment;

(3) FIG. 2 is a partial schematic view of one example of a turbomachine which can comprise a rotating assembly in accordance with an embodiment;

(4) FIG. 3 schematically illustrates an aircraft comprising two turbomachines in accordance with one embodiment.

(5) In all the figures, similar elements bear identical references.

DETAILED DESCRIPTION

(6) A turbofan engine **1**, in particular of an aircraft **100**, generally comprises, from upstream to downstream in the gas flow direction, a fan **2** (or propeller), an annular primary flow space and an annular secondary flow space which is externally relative to the primary flow. The mass of air sucked in by the fan **2** is therefore divided into a primary stream which circulates in a primary flowpath, and a secondary stream which is concentric with the primary stream and circulates in a secondary flowpath. The fan **2** (or propeller) can be ducted and housed in a fan **2** casing or in an unducted variant of the USF (Unducted Single Fan **2**) type. The fan blades **2** can be fixed or have variable setting, the setting being adjusted according to the flight phases by a pitch change mechanism.

(7) The primary flowpath passes through a primary body comprising one or several stages of compressors, for example a low-pressure compressor **3** (or booster) and a high-pressure compressor **4**, a combustion chamber, one or several stages of turbines, for example a high-pressure turbine **5** and a low-pressure turbine **6**, and a gas exhaust nozzle. Typically, the high-pressure turbine **5** drives in rotation the high-pressure compressor **4** via a first shaft **7** called high-pressure shaft, while the low-pressure turbine **6** drives in rotation the low-pressure compressor **3** and the fan **2** via a second shaft **8** called low-pressure shaft. The low-pressure shaft **8** is generally housed on a certain section in the high-pressure shaft **7**.

(8) Several components **11**, **15**, **29** of the turbomachine **1** can be removably mounted on the low-pressure shaft **8** in order to be driven in rotation about the axis of revolution of the low-pressure shaft **8**. These components **11**, **15**, **29** comprise in particular one or several bearings **11**, sealing elements **15**, **16**, **30** and/or one or several scoops **29**.

(9) As described above, a bearing **11** is configured to support a mobile part relative to a fixed part (or stator) of the turbomachine **1** or relative to another mobile part. The bearing **11** comprises in particular an inner ring **12** secured in movement to the low-pressure shaft **8**, an outer ring **13** mounted on another part of the turbomachine **1**, the inner ring **12** and the outer ring **13** housing for example a rolling **14**. The inner ring **12** and the outer ring **13** are mobile relative to each other.

(10) The sealing element **15** is configured to achieve a sealing between two pieces and can comprise for example a labyrinth seal which has at least two axially adjacent clearance restrictions. The clearance restrictions can for example comprise at least two knife-edges and/or a twist which winds on at least two turns. The labyrinth seal can thus comprise a support secured in movement to

the low-pressure shaft **8** and the clearance restrictions configured to cooperate with a complementary sealing piece mounted on a facing piece, for example an abradable element. It will be noted that the sealing element **15** can comprise any other type of seal, such as a segmented radial seal, a brush seal, a carbon seal, etc.

(11) The scoop **29** can be part of a lubrication system of the rotating assembly **1** and be configured to supply pieces with lubrication fluid.

(12) In order to facilitate the manufacture and maintenance of the low-pressure shaft **8** and to improve its service life, all or part of the components **11**, **15**, **16**, **29** driven in rotation by the low-pressure shaft **8** are connected to the shaft via a sleeve **19** fitted onto the low-pressure shaft **8**. For this purpose, the low-pressure shaft **8** comprises a first flange **18** extending radially from an outer radial surface of the shaft. Moreover, the sleeve **19** comprises a second flange **20** extending radially from the sleeve **19** and configured to cooperate with the first flange **18** in order to allow the sleeve **19** to be fixed on the low-pressure shaft **8**. The sleeve **19** is then secured in movement (in rotation and in translation) to the low-pressure shaft. The components **11**, **15**, **17**, **29** are then mounted on or formed in the sleeve **19** and driven by the low-pressure shaft **8** via the sleeve **19**. Particularly, the components **11**, **15** and **29** are not directly mounted on the low-pressure shaft **8** and are preferably not in direct contact with the low-pressure shaft **8**.

(13) The first flange **18** is monolithic with the low-pressure shaft **8** in order to limit the machining needs of the shaft. The second flange **20** can be monolithic with the sleeve **19** or added and fixed onto the sleeve **19**. The sleeve **19** and the low-pressure shaft **8** form a rotating assembly **10**. The first flange **18** and the second flange **20** extend here radially outwards relative to the rotation shaft X of the low-pressure shaft **8**.

(14) Removably fixing the components **11**, **15**, **29** on the sleeve **19**, rather than on the low-pressure shaft **8**, makes it possible to limit the machining of the low-pressure shaft **8** which improves its service life. It is further simpler and less expensive to replace or repair worn or damaged components **11**, **15**, **29**, since it suffices to dismount the components **11**, **15** and/or the sleeve **19** in order to replace them. Fixing the sleeve **19** via a flange rather than any other connection means also avoids making threads prone to damage during the operation of the turbomachine **1**, and can be made with greater manufacturing and alignment tolerance.

(15) The rotating assembly **10** further comprises a lubrication enclosure **25** comprising a wall **25a** fixedly mounted on a stator of the turbomachine, for example a casing of the turbomachine. The bearing **11** is housed in the lubrication enclosure **25**. The inner ring **12** is added and fixed onto the sleeve **19** and secured in rotation to the sleeve **19**, and an outer ring **13** fixedly mounted relative to the wall **25a** of the lubrication enclosure **25**.

(16) The enclosure **25** is delimited on either side by sealing elements **15**, **16** comprising clearance restrictions. Advantageously, the use of clearance restrictions such as knife-edges or twists rather than cylindrical seals improves the sealing of the enclosure **25**, the sealing being achieved over a greater axial length. In addition, possible damage to one of the clearance restrictions is compensated by the presence of the other clearance restrictions of the sealing element **15**, **16** extending immediately downstream, so that the sealing of the enclosure **25** is more robust. However, in the case of high bypass ratio turbomachines **1**, the rotation speed of the low-pressure shaft **8** is very high (the low-pressure turbine being a fast turbine), so that it is necessary to guarantee the sealing of the enclosure **25** housing the bearings **11** mounted on the low-pressure shaft **8**.

(17) A first **15** of the sealing elements comprises a first part **15a** added and fixed onto the sleeve **19** while being secured in movement to the sleeve **19** and a second part **15b** fixedly mounted relative to the wall **25a** of the enclosure **25**. A second **16** of the sealing elements comprises a first part **16a** fixedly mounted on the sleeve **19** while being secured in movement to the sleeve **19** and a second part **16b** fixedly mounted relative to the wall **25a** of the enclosure **25**. In the exemplary embodiment of FIG. **1**, the first part **16a** is added and fixed onto the sleeve **19**. As a variant, the

first part **16** can be monolithic with the sleeve **19**.

(18) The bearing **11** is placed between the first and the second sealing element **15**, **16**.

(19) The sleeve **19** further comprises a shoulder **23** extending at a distance from the second flange **20**, the first sealing element **15** and the inner ring **12** being tightly mounted on the sleeve between a nut **22** and the shoulder **23**. This configuration allows making the sealing element **15** and the inner ring **12** remobile relative to the sleeve **19** while guaranteeing their axial positioning on the sleeve **19**. The mounting of these components **15**, **12** is further easy since it is sufficient to fit them onto the sleeve **19** by putting them in abutment against the shoulder **23** and to block them in position using the nut. The sleeve **19** is further easy to make, for example by machining a thread at a free end of the sleeve **19**, the thread being configured to cooperate with threads of complementary shape and pitch of the nut **22**, thus allowing the tightening of the components **12**, **15** against the shoulder **23** by simply screwing the nut **22**.

(20) The shoulder **23** can be monolithic with the sleeve **19**. As a variant, the shoulder **23** can be formed by a piece added and fixed onto the sleeve **19**, for example a threaded nut screwed onto the sleeve **19**, or a mechanically fixed or welded flange. The shoulder **23** extends radially from the sleeve **19**. All or part of the components **15**, **11**, **29** are placed in abutment, directly or indirectly, against the sleeve **19** and blocked in position using the nut **22**. The nut applies the components **15**, **11**, **29** in series against the shoulder **23**. The nut **22** and the sleeve **19** can for example comprise complementary threads.

(21) When the first part **16a** of the second sealing element is added onto the sleeve **19** (see FIG. 1 for example), this first part **16a** is also preferably tightly mounted on the sleeve **19** between the nut **22** and the shoulder **23**. Typically, the first part **16a** can be positioned axially between the inner ring **12** of the bearing **11** and the shoulder **23**.

(22) On the other hand, when the first part **16a** of the second sealing element **16** is monolithic with the sleeve **19**, the shoulder **23** can be placed either between the inner ring **12** and the first part **16**, or between the inner ring **12** and the first sealing element **15**. In this case, the inner ring can for example comprise an axial portion **12a** which supports the rolling of the bearing **11** and a radial portion **12b** which extends from an upstream edge or a downstream edge of the axial portion.

(23) In one embodiment, the shoulder **23** extends at a distance from the second flange **20** and the nut **22** is mounted on the free end of the sleeve **19** extending opposite to the second flange **20** relative to the shoulder **23**. The sleeve **19** therefore comprises, from its free end, the nut **22**, the first sealing element **15**, the bearing **11**, the second sealing element **16** and the second flange **20**.

(24) The second flange **20** is placed in abutment against the first flange **18** and held in this position, for example using a bolt **21**. For this purpose, the first flange **18** and the second flange **20** each comprise at least one orifice, and a fixing member **21** is inserted into these orifices in order to connect the flanges **18**, **20**. The orifices can be through and smooth orifices, in which case the fixing member **21** can comprise a threaded rod and a nut. As a variant, at least one of the orifices can be threaded, in which case the fixing member **21** can comprise a screw and optionally a nut. Other fixing members can of course be envisaged to connect the first flange **18** and the second flange **20**.

(25) The fixing member **21** extends radially outward relative to the sleeve **19**. The sleeve **19** is therefore closer to the axis X than the fixing member **21**, which contributes to the sealing of the lubrication enclosure.

(26) The components **15**, **16**, **11**, **29** mounted on or formed in the sleeve **19** can comprise at least the inner ring **12** of the bearing **11**, the first and second sealing elements **15**, **16** and optionally at least one scoop **29**. In the illustrated exemplary embodiment, the bearing **11** is placed between the two sealing elements **15**, **16**, which axially delimit the lubrication enclosure **25**. A scoop **29** is further interposed between the first sealing element **15** and the bearing **11**. The first sealing element **15**, the scoop **29**, the bearing **11** and where appropriate the second sealing element **16** are tightly mounted in series between the nut **22** and the shoulder **23** of the sleeve **19**. Neither the nut **22**, nor

the components **11**, **15**, **16**, **29** are therefore fixed, nor even in direct contact, with the low-pressure shaft **8**.

(27) Here, the first sealing element **15** comprises a tubular support **15c** mounted on the sleeve **19** and a shroud **15a** extending upstream from the tubular support **15c** and carrying clearance restrictions. The shroud **15a** extends radially around the nut **22**. The second sealing element **16** further downstream also comprises a support **16c**, which is either connected to the sleeve **19** between the shoulder **23** and the inner ring **12** of the bearing **11** (FIG. 1), or monolithic with the sleeve **19**, and a shroud **16a** extending radially around the sleeve **19** between the support **16c** and the second flange **20** and which carries the clearance restrictions.

(28) In one embodiment, the clearance restrictions of the second sealing element **16** comprise a first group of clearance restrictions mounted on the shroud **16a** and a second group of clearance restrictions mounted on the part **16b** fixed onto the wall **25a** of the enclosure. The two groups of clearance restrictions are spaced axially so that the clearance restrictions of the first group face an abradable fixed onto the part **16b** while the clearance restrictions of the second group face an abradable fixed onto the shroud **16a**. This configuration makes it possible to improve the sealing of the enclosure **25**, the axial length of the sealing element **16** being greater due to the presence of the clearance restriction groups and to the space separating them.

(29) In one embodiment, a sum of an axial length of the first sealing element **15** and of the second sealing element **16** is at most equal to twice an axial distance between the nut **22** and the shoulder **23**, preferably substantially equal to within 10%.

(30) In order to further improve the sealing of the rotating assembly **10**, the assembly **10** further comprises a third sealing element **30** comprising clearance restrictions (knife-edges or twists), the third sealing element **30** comprising a first part **30a** added and fixed onto the first flange **18** by being secured in movement to the low-pressure shaft **8** and a second part **30b** fixedly mounted relative to the wall **25a** of the enclosure **25**. Where appropriate, the first part **30a** of the third sealing element **30** can be fixed onto the first flange **18** using the fixing member **21**, for example between the first and the second flange **18**, **20** when the fixing member **21** comprises a bolt. Preferably, the bolt **21** is radially outside relative to the sleeve **19**.

(31) Still to improve the sealing of the rotating assembly **10**, the third sealing element **30** is outside the enclosure **25**, unlike the first and second sealing elements **15**, **16** which axially delimit the enclosure **25** and allow the relative rotation of the wall **25a** relative to the low-pressure shaft **8**. For this, the second part **30b** of the third sealing element **30** can be formed by a cylindrical wall extending axially from the wall **25a** of the enclosure **25** in the direction of the second flange **20**, while the first part **30a** is bent and comprises a radial portion mounted on the first flange **18** and an axial portion extending in the direction of the enclosure **25** so as to face the cylindrical wall **30b**.

(32) The third sealing element **30** moreover extends radially outward relative to the second sealing element **16**.

(33) Where appropriate, the first flange **18** of the low-pressure shaft **8** can be connected to an additional piece **24** of the turbomachine **1**, for example to a flange of a rotor (as illustrated in FIG. 1). Where appropriate, the flange **24** extends radially outward relative to the fixing member **21**.

(34) The turbomachine **1** can also comprise a lubrication system configured to supply pieces with lubrication fluid, for example the bearing **11**. For example, the lubrication system comprises a nozzle **26** opening out into the enclosure **25** and configured to provide an amount of fluid (typically oil) to the bearing **11**. In the example illustrated in the figures, the enclosure **25** comprises the bearing **11** and is delimited upstream and downstream by the sealing elements **15** mounted on the sleeve **19**. The nozzle **26** can in particular be positioned between the bearing **11** and the most upstream sealing element. The scoop **29** is then mounted on the sleeve **19** so as to extend between the nozzle **26** and the low-pressure shaft **8**. It is configured to convey the fluid provided by the nozzle **26** up to the rolling **14** of the bearing **11**. For this purpose, the scoop **29** can in particular comprise a sheath fitted onto the sleeve **19** in which are formed an orifice **17** and a supply channel

27 connecting the orifice 17 to the inner ring 12. When the scoop 29 is fixed onto the sleeve 19, the orifice 17 is located in the vicinity of the nozzle 26, for example facing the nozzle 26. The supply channel 27 is preferably formed in the scoop 29 and extends axially from the orifice 17 up to one or several ducts 28 formed in the inner ring 12 and opening out onto the rolling 14. Under the effect of the centrifugal forces, the fluid is then projected by the ducts 28 onto the rolling 14 and diffuses into the enclosure 25.

(35) In one embodiment, the wall 25a of the lubrication enclosure 25 comprises two parts each comprising a fixing flange 31a, 31b. The fixing flanges 31a, 31b of the wall 25a are both fixed onto the same flange 32 of the stator of the turbomachine via a bolt 33. For example, the fixing flanges 31a, 31b can be mounted from on either side of the flange 32 of the stator, the three flanges 31a, 32, 31b being fixed together by the bolt 33. Preferably, each fixing flange 31a, 31b is monolithic with the part of the corresponding wall 25a.

(36) The present disclosure finds a particular application in the case of a turbomachine 1 of the USF type, whose propeller (or fan 2) is unducted and is driven by the low-pressure shaft 8 via a reduction gear 9. The propeller then has a high bypass ratio, typically a bypass ratio greater than 30, for example between 40 and 80. By bypass ratio, it will be understood here the ratio between the flow rate of the secondary stream and the flow rate of the primary stream. The low-pressure shaft 8 of such a turbomachine 1 is then supercritical (that is to say it presents a bending deformation mode in the operating range) and fast. It is therefore subject to a severe vibrational environment. In addition, the materials usually used for these low-pressure shafts 8 can be susceptible to corrosion and wear.

(37) Fixing in particular the sealing elements 15 on the sleeve 19 rather than on the low-pressure shaft 8 thus allows making the sealing elements 15 in materials distinct from the material of the low-pressure shaft 8, and thus reducing the risks of corrosion, in particular at the head of clearance restrictions. The maintenance is further greatly simplified since it is sufficient to dismount the sealing element 15 from the sleeve 19 or possibly the sleeve 19 from the low-pressure shaft 8.

Claims

1. An assembly of a turbomachine comprising: a shaft (8) configured to drive a propeller of the turbomachine about an axis and comprising a first flange extending radially from the shaft, the first flange being monolithic with the shaft; a sleeve fitted onto the shaft and comprising a second flange extending radially from the sleeve, the sleeve being fixed onto the first flange of the shaft via the second flange; a lubrication enclosure comprising a wall configured to be fixedly mounted on a stator of the turbomachine; a bearing housed in the lubrication enclosure, the bearing comprising an inner ring mounted on the sleeve and secured in rotation with respect to the sleeve, and an outer ring configured to be fixedly mounted relative to the wall of the lubrication enclosure; a first sealing element comprising a first labyrinth seal which has two axially adjacent clearance restrictions, the first sealing element comprising a first part added and fixed onto the sleeve) by being secured in movement with the sleeve and a second part fixedly mounted relative to the wall of the lubrication enclosure; a second sealing element comprising a second labyrinth seal which has two axially adjacent clearance restrictions, the second sealing element comprising a first part fixedly mounted on the sleeve by being secured in movement to the sleeve and a second part fixedly mounted relative to the wall of the lubrication enclosure, the bearing being placed axially between the first and the second sealing element; the sleeve further comprising a shoulder extending axially at a distance from the second flange, the first sealing element and the inner ring being tightly mounted axially on the sleeve between a nut and the shoulder.

2. The assembly according to claim 1, wherein the sleeve is made of a material distinct from the material of the shaft.

3. The assembly according to claim 1, wherein the clearance restrictions of at least one of the first

and second labyrinth seals comprise at least two knife-edges or a twist that winds on at least two turns.

4. The assembly according to claim 1, further comprising a third sealing element comprising axially adjacent clearance restrictions, the third sealing element comprising a first part added and fixed onto the first flange by being secured in movement to the shaft and a second part fixedly mounted relative to the wall of the lubrication enclosure.

5. The assembly according to claim 4, wherein the third sealing element is disposed radially outside relative to the second sealing element.

6. The assembly according to claim 4, wherein the third sealing element is placed outside the lubrication enclosure.

7. The assembly according to claim 1, wherein the first sealing element and the inner ring are removably mounted on the sleeve.

8. The assembly according to claim 7, further comprising a third sealing element comprising axially adjacent clearance restrictions, the third sealing element comprising a first part added and fixed onto the first flange by being secured in movement to the shaft and a second part fixedly mounted relative to the wall of the lubrication enclosure, wherein one of the second and the third sealing element is also removably mounted on the sleeve.

9. The assembly according to claim 1, wherein the first part of the second sealing element is monolithic with the sleeve.

10. The assembly according to claim 1, wherein the second flange is monolithic with the sleeve.

11. The rotating assembly according to claim 1, wherein the nut is screwed on a free axial end of the sleeve.

12. The assembly according to claim 1, further comprising an additional flange, the additional flange, the second flange and the first flange being connected by a same fixing member.

13. The assembly according to claim 12, wherein the fixing member is a bolt, which is arranged radially outside relative to the sleeve.

14. The assembly according to claim 1, wherein the wall of the lubrication enclosure comprises a third flange and the stator of the turbomachine comprises a fourth flange, the third flange being configured to be fixed onto the fourth flange via a second bolt.

15. The assembly according to claim 14, wherein the wall of the lubrication enclosure comprises two wall portions, a first of the wall portions being monolithic with the third flange and a second of the wall portions being monolithic with a fifth flange, the fifth flange also being fixed onto the fourth flange via the second bolt.

16. The assembly according to claim 15, further comprising an additional flange, the additional flange, the second flange and the first flange being connected by a same fixing member, wherein the fixing member is a bolt, which is arranged radially outside relative to the sleeve, the additional bolt being arranged radially outside the fixing member.

17. The assembly according to claim 1, wherein the two axially adjacent clearance restrictions of the second labyrinth seal comprise a first group of clearance restrictions mounted on the first part and a second group of clearance restrictions mounted on the second part, the first group of clearance restrictions being spaced axially from the second group of clearance restrictions.

18. The assembly according to claim 1, wherein a sum of an axial length of the first sealing element and of the second sealing element is at most equal to a double of an axial distance between the nut and the shoulder.

19. The assembly according to claim 1, wherein the first sealing element comprises a first support fixed onto the sleeve and a first shroud extending radially around the nut, one among the first support and the first shroud having the clearance restrictions of the first labyrinth seal.

20. The assembly according to claim 1, wherein the second sealing element comprises a second support fixedly mounted on the sleeve and a second shroud, the second shroud extending axially between the bearing and the second flange.

21. The assembly according to claim 1, further comprising: a nozzle opening out into the lubrication enclosure, the nozzle being connected to a lubrication system and being configured to provide an amount of fluid to the bearing; and a scoop connected to the lubrication system and configured to convey the fluid to the bearing.
22. The assembly according to claim 21, wherein the scoop has an inlet arranged facing the nozzle and an outlet which opens out onto the inner ring.
23. The assembly according to claim 21, wherein the scoop comprises a sheath fitted onto the sleeve in which are formed an orifice and a supply channel fluidly connecting the inlet and the inner ring of the bearing.
24. The assembly according to claim 23, wherein the supply channel extends axially between the inlet and the outlet.
25. A turbomachine comprising the assembly according to claim 1 and an unducted propeller driven in rotation by the shaft, the turbomachine having a bypass ratio greater than 30.
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